

Entrapment Strategies to Enhance the Bioavailability of Quercetin: A Literature Review

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BMEG 210 Term Project

Abstract

Quercetin is a plant flavonoid with antidiabetic, anti-inflammatory and antitumour activity, but poor aqueous solubility, chemical instability and low absorption limit its clinical use. This review examines four carrier strategies for entrapping quercetin: emulsions, hydrogels, liposomes and polymeric micelles. For each, representative studies are summarised in terms of entrapment efficiency, stability and reported bioavailability gains. Emulsions and hydrogels achieve high entrapment and good storage stability, liposomes are vulnerable to intestinal phospholipases, and polymeric micelles show the largest pharmacokinetic gains in the most thoroughly characterised in vivo study, together with options for active targeting. Differences in metrics, models and administration routes limit direct comparison, and human trials are still lacking.

Keywords: quercetin; bioavailability; encapsulation; nanoemulsion; hydrogel; liposome; polymeric micelle; type 2 diabetes

1. Introduction

Diabetes mellitus is characterised by chronically high blood glucose arising from insufficient insulin secretion or insulin resistance, and is among the most prevalent metabolic disorders worldwide [1]. Interest in natural compounds as adjunct therapy has grown, and quercetin has emerged as a candidate because of its glucose-regulating properties [4]. Its clinical application, however, is limited by poor solubility and bioavailability. This review evaluates delivery systems that entrap quercetin and asks which is the most promising.

1.1 Diabetes

Diabetes has several subtypes, including type 1, type 2, maturity-onset, gestational, neonatal and steroid-induced diabetes, each with a distinct pathophysiology and management. Type 1 diabetes mellitus (T1DM) is an autoimmune disease that destroys insulin-producing pancreatic β cells, causing severe insulin deficiency; it usually presents in childhood or adolescence and requires lifelong insulin therapy [2]. Type 2 diabetes

mellitus (T2DM) is marked by insulin resistance in peripheral tissues and a progressive decline in β -cell function [3]. It develops gradually, is associated with obesity, poor diet and inactivity, and can often be managed with oral antidiabetic drugs [3][4].

Both forms involve the pancreas, where β cells secrete insulin and α cells secrete glucagon. These cells continuously adjust hormone output to the glucose environment, and when this balance fails, blood glucose becomes inappropriately elevated [1].

1.2 Quercetin and the bioavailability problem

Quercetin is a naturally occurring flavonoid found in many plants and vegetables, with reported antidiabetic, anti-inflammatory and antitumour effects [5]. Bioavailability, the fraction of an ingested substance that is absorbed and available for physiological activity, is its main limitation. Quercetin has poor water solubility (about 60 mg/L at 16 °C) and is sensitive to pH, temperature and co-ingested food components; orally administered quercetin is reported to be only about 6.7% absorbed [6]. Overly polar forms such as glycosides may be unable to cross epithelial phospholipid bilayers passively, and glycosides are hydrolysed to aglycones by bacterial enzymes in the lower intestine [7]. These constraints motivate carrier systems that protect and solubilise quercetin [6][8].

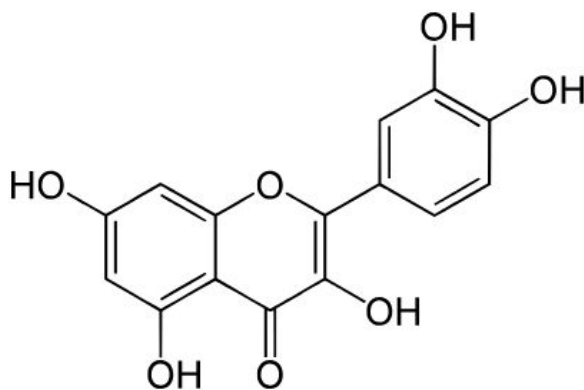


Figure 1. Chemical structure of quercetin.

2. Delivery Systems for Quercetin

Encapsulation traps an active agent within another material to improve its handling. Four families of carrier are considered below: emulsions, hydrogels, liposomes and polymeric micelles.

2.1 Emulsions

Emulsions are thermodynamically unstable colloidal dispersions of two or more immiscible liquids. In an oil-in-water emulsion, emulsifier-coated oil droplets are dispersed in water and can host hydrophobic quercetin. Droplets larger than 200 nm are conventional emulsions and those below 200 nm are nanoemulsions; below about 50 nm they can be optically transparent, and rapid digestion of small droplets in the gastrointestinal

tract can improve bioavailability. Variants include Pickering, multilayer, multiple and high internal phase emulsions (HIPEs), all designed to improve the encapsulation, protection and delivery of hydrophobic bioactives.

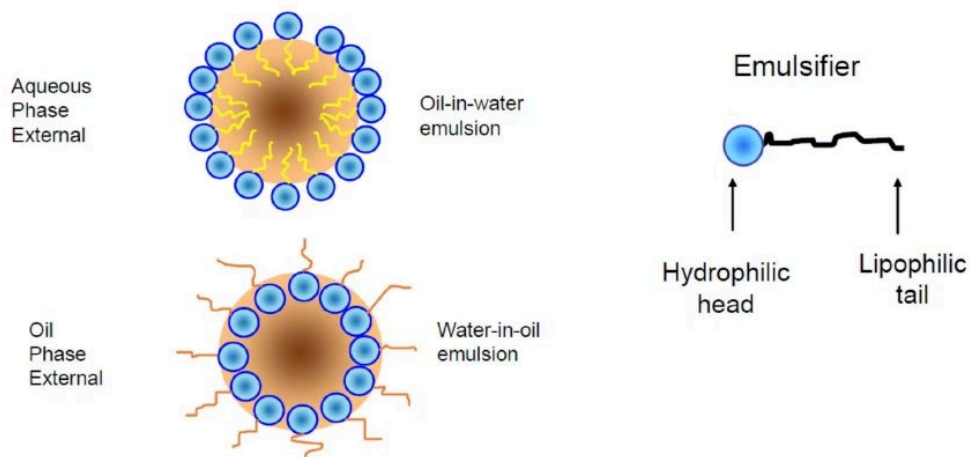


Figure 2. An emulsion droplet stabilised by surfactant molecules at the interface between internal and external phases. Reproduced from [9].

Ultrasonic processing. Soy protein isolate was used to stabilise an olive-oil-in-water emulsion. Quercetin entrapment efficiency reached 77.39% at lower pH, emulsion stability improved as pH decreased, and the highest reported value, 15.49%, was observed at pH 7 [10].

Ultra-high-pressure homogenisation. A rice-protein nanoemulsion showed higher encapsulation efficiency as quercetin content increased, greater stability under non-acidic conditions and low salt concentration, and a 9-fold improvement in bioaccessibility [6].

2.2 Hydrogels

Hydrogels are built by physically or chemically cross-linking polysaccharides. Quercetin can be held by strong attractive interactions with the polymer network or within hydrophobic domains of colloidal particles such as lipid droplets or protein nanoparticles; in the latter case, loaded particles must be larger than the pore size of the network [6]. Mirpoor et al. entrapped more than 93% of quercetin [11].

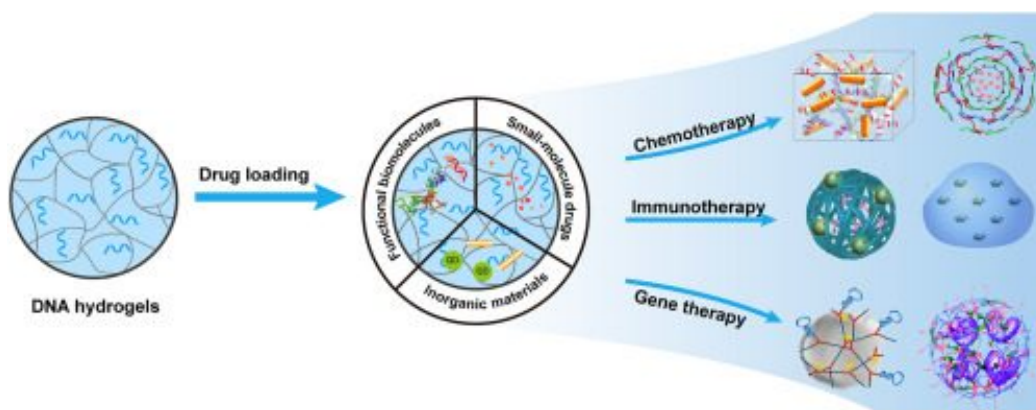


Figure 3. Drug-loading applications of hydrogels (DNA hydrogel example).

Injection under magnetic stirring. Whey protein isolate coated with lotus-root amylopectin gave 92.4% encapsulation efficiency at pH 7, with almost 80% of quercetin retained after 35 days of storage. Mice fed these hydrogels showed eight times more quercetin in serum [21].

Ionic gelation. Hydrogel beads of oligochitosan and de-esterified yuzu-peel pectin stayed roughly intact through in vitro digestion until pectinase was added in simulated colonic fluid, indicating potential for colon-targeted release [12].

2.3 Liposomes

Liposomes are colloidal suspensions of phospholipid vesicles. Phospholipids have a hydrophilic head and two hydrophobic tails, so in water they form a bilayer with the heads outward; hydrophobic quercetin partitions into the bilayer core [6][13]. Liposomes can protect cargo from reactive enzymes in the stomach [14], but phospholipases in the small intestine degrade the phospholipids [15]. Loading quercetin-containing liposomes into chitosan hydrogel beads improved liposome stability during simulated gastrointestinal digestion and slowed release of the bioactives [22].

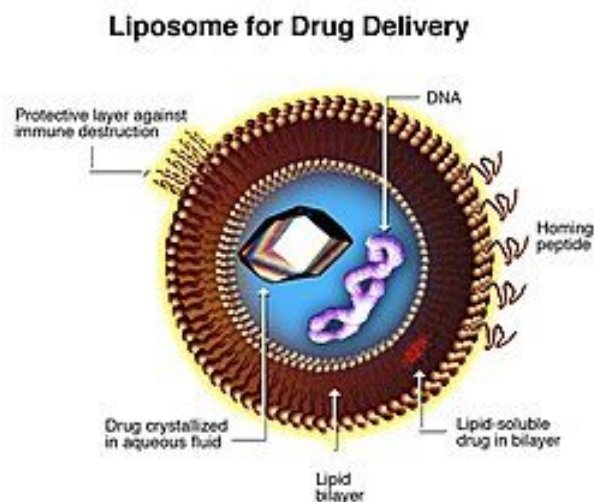


Figure 4. Liposome structure for drug delivery. Reproduced from [16].

2.4 Polymeric micelles

Polymeric micelles are nanoscale particles formed from amphiphilic polymers in water, with a hydrophilic shell and a hydrophobic core that solubilises non-polar drugs such as quercetin. They can improve bioavailability, stability and solubility and can be modified for targeted delivery [17].

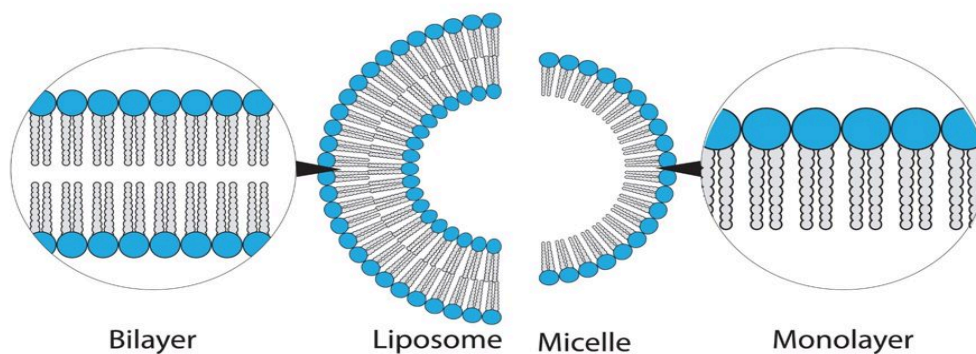


Figure 5. Bilayer, liposome, micelle and monolayer architectures compared.

Chang et al. made lecithin-stabilised polymeric micelles by thin-film hydration followed by ultrasonication, using TPGS and lecithin. Relative to free quercetin, bioavailability rose by 158% after oral and 360% after intravenous administration. Mean particle size was 92.2 nm, encapsulation efficiency 98.88% and drug loading 4.72% [18].

Xu et al. developed a hyaluronic acid–quercetin (HA–QU) conjugate micelle that carries docetaxel and overcomes multidrug resistance. The amphiphilic conjugate was made by esterification in anhydrous formamide with EDC and NHS to activate the carboxyl groups of hyaluronic acid. Particles measured 176.8 ± 3.4 nm, with entrapment efficiency $86.8 \pm 1.5\%$ and drug loading $23.6 \pm 1.3\%$ [19]. Beyond passive carriage, hyaluronic acid ligands bind CD44 receptors and so target disease sites and increase intracellular drug efficacy.

3. Comparative Analysis

High entrapment efficiency does not guarantee high bioavailability: entrapment efficiency is the fraction of drug held by the carrier, whereas bioavailability is the fraction the body absorbs and uses. Table 1 collects the reported figures.

Table 1. Reported performance of quercetin delivery systems (values as given by the cited studies).

System	Formulation	Reported result	Ref.
Emulsion	Soy protein isolate, olive oil, ultrasonic	Entrapment 77.39%; 15.49% at pH 7	[10]
Nanoemulsion	Rice protein, ultra-high-pressure homogenisation	9-fold bioaccessibility	[6]
Hydrogel	β -lactoglobulin core–shell nanostructure	>93% entrapped	[11]
Hydrogel	Whey protein isolate + lotus-root amylopectin	Entrapment 92.4%; ~80% retained at 35 d; 8× serum quercetin in mice	[21]
Hydrogel beads	Oligochitosan + yuzu pectin, ionic gelation	Intact until pectinase in simulated colonic fluid	[12]
Liposome	Liposomes in chitosan hydrogel beads	Improved liposome stability in simulated GI digestion; slower release	[22]
Polymeric micelle	Lecithin-stabilised (TPGS, lecithin)	92.2 nm; entrapment 98.88%; +158% oral, +360% IV	[18]
Polymeric micelle	Hyaluronic acid–quercetin conjugate	176.8 nm; entrapment 86.8%; loading 23.6%; CD44 targeting	[19]

4. Discussion

Micelles combine high entrapment with measured pharmacokinetic gains and the option of ligand-based targeting, whereas liposomes are limited by phospholipase degradation in the small intestine. Several caveats apply.

Metrics are not comparable. The studies report entrapment efficiency, bioaccessibility (in vitro) and bioavailability (in vivo), in different species, at different pH values and by different routes. For example, the 8-fold rise in mouse serum quercetin for the hydrogel and the 9-fold bioaccessibility gain for the

nanoemulsion are larger in headline size than the 158% oral increase for micelles, yet they measure different things. A definitive ranking would require a head-to-head study with a common model and endpoint.

Relevance to diabetes. The micelle studies were conducted in cancer settings, and intravenous delivery is less relevant to a chronic oral therapy for T2DM.

Evidence base. Most work is in vitro or in animals, so controlled human trials are needed to establish dosage and efficacy across demographics. Findings are also sometimes inconsistent; for instance, reports conflict on whether quercetin activates or inhibits PPAR γ , a regulator of obesity [20].

Quercetin's low toxicity and broad activity suggest that advanced carriers could offer a safer, natural adjunct therapy for T2DM and metabolic syndrome [4]. Nanoformulations such as liposomes and nanoemulsions are already being tested for functional foods and pharmaceutical forms, including hydrogels for wound healing and topical delivery [6].

5. Conclusion

Emulsions and hydrogels entrap quercetin efficiently and protect it in storage, and liposomes are undermined by intestinal phospholipases. Polymeric micelles emerge as a leading option because they pair high entrapment with reported bioavailability gains of 158% (oral) and 360% (intravenous) [18] and can be conjugated for active cellular targeting [19]. Future work should compare these carriers under common conditions and move from preclinical formulations to human trials.

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